BIOPHARMACEUTICS & PHARMACOKINETICS

UNIT 2 NOTES

ELIMINATION

- DRUG METABOLISM
- RENAL EXCRETION OF DRUGS

BIOAVAILABILITY

- DEFINITION & OBJECTIVES
- MEASUREMENT
- BIOEQUIVALENCE STUDIES

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ELIMINATION

Elimination is the process by which a drug is removed from the systemic circulation either unchanged or after being metabolized, resulting in the termination of its pharmacological effects.

COMPONENTS OF ELIMINATION

① The two major components of Elimination are :
② Metabolism (Biotransformation)
Excretion.

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ⓘ **Metabolism**
- It occurs mainly in liver.
- Also known as Biotransformation.
- It is the process by which Active drug is converted into excreted form.

ⓘⓘ **Excretion**
- It mainly occurs in kidneys.
- It is the process by which drug & its metabolites are removed from the body.

METABOLISM

- It is also known as 'Biotransformation'.
- It is a process by which body transforms a drug into more readily excretable forms.
- The primary site for metabolism is Liver.
- Other sites include kidney, intestines, lungs and plasma
- Metabolism usually involves enzymatic reactions in the liver that alter (changes) the chemical structure of drug.
- These reactions makes the drug more water soluble and thus drugs becomes easier to be eliminated from the body via urine or bile.
- The metabolism of a drug usually converts:
  - Lipid Soluble → Water Soluble
  - Unionised → Ionized
- Metabolism is crucial for drug's duration and intensity of action, as well as for its overall safety profile.

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DRUG METABOLISM

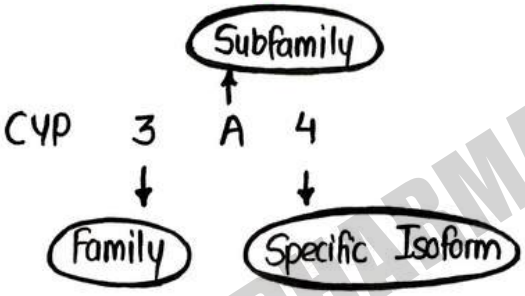
Metabolism usually leads to conversion of:

- Active Drug → Active Metabolite
- Active Drug → Inactive Metabolite
- Inactive Drug → Active Metabolite

Cytochrome P450 Enzymes

- Cytochrome P450 enzymes are group of proteins in the body that help breakdown drugs, toxins and other substances
- In Cytochrome P450, P stands for pigment that has maximum light absorption at wavelength 450 nm.
- Several families of CYP enzymes are involved in metabolism of drugs.
- These are named as CYP followed by a number (denotes family), then alphabet (subfamily) and again a number (specific isoform of the enzyme)

Example :



- CYP 3A4 forms the maximum hepatic content (26%) of CYP enzymes and is involved in metabolism of maximum percentage of drugs (33%).

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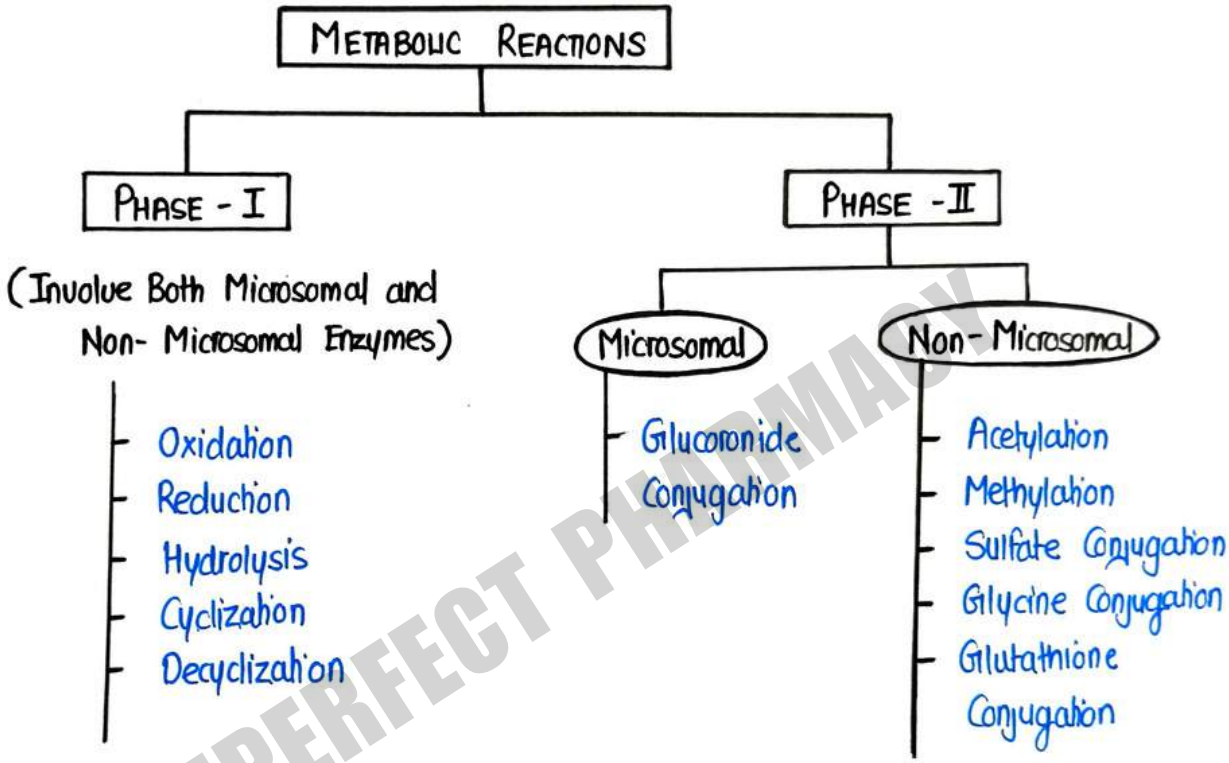
Example Of CYP Enzymes

- CYP3A4
- CYP2D6
- CYP2C19
- CYP2C9
- CYP1A2

Types Of Metabolism Reaction

There are mainly two types of Biotransformation reactions :

- Phase I Reaction
- Phase II Reaction



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① Phase - I Reactions

These reactions introduce functional groups in the drug. It includes :

- Oxidation
- Reduction
- Hydrolysis
- Cyclization
- Decyclization

ⓐ Oxidation

- Oxidation is the process of addition of oxygen to a drug molecule or removal of hydrogen from a drug molecule.
- Oxidation is primarily mediated by enzymes like CYP450s and results in the formation of more polar metabolites, which can be easily excreted from the body.
- Example : Phenytoin, Phenobarbitone, Propranolol

ⓑ Reduction

- Reduction is the process of addition of Hydrogen or removal of Oxygen from a drug molecule.
- It is less common than Oxidation.
- Example : Chloramphenicol, Warfarin etc.

ⓒ Hydrolysis

- Breakdown of drug molecule by addition of water is known as Hydrolysis.
- This is common among esters and amides.
- It is mediated by enzymes such as esterase, amidases & pephidases.

ⓓ Cyclization

- In this, straight chain compound converted into ring structure.
- example : Cycloguanil from Proguanil

ⓔ Decyclization

- It involves opening up of ring structure of cyclic drug molecule.
- Example : Barbiturates.

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② PHASE II REACTIONS

- Phase II Reactions are also known as Conjugation Reactions
- These reactions attach polar groups (e.g. glucuronic acid, sulfate, acetyl, methyl) to the drug making it highly water soluble.
- These reactions include :

ⓐ Glucuronide Conjugation
ⓑ Acetylation
ⓒ Methylation
ⓓ Sulfate Conjugation
ⓔ Glycine Conjugation
ⓕ Glutathione Conjugation

ⓐ Glucuronide Conjugation

- It is the process of addition of Glucuronic Acid.
- It is mediated by UDP - glucuronosyl transferases (UGTs)
- Examples : Chloramphenicol, aspirin, paracetamol etc.

ⓑ Acetylation

- It is the process of addition of Acetyl Group.
- It is mediated by N- Acetyltransferases.
- Example : Sulfonamides, isoniazid.

ⓒ Methylation

- It is the process of addition of Methyl Group.
- It is mediated by Methyltransferases.
- Example : Methyldopa, Mercaptopurine.

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ⓓ Sulfate Conjugation

- It is the process of addition of Sulfate Group.
- It is mediated by Sulfotransferases.
- Example: Chloramphenicol, Methyldopa etc.

ⓔ Glycine Conjugation

- It is process of addition of Glycine to the drugs.
- It is mediated by Glycine- N- Acetyltransferases.
- Example: Salicylates, Nicotinic Acid etc.

ⓕ Glutathione Conjugation

- It is the process of addition of Glutathione.
- It is mediated by Glutathione - S- Transferases.
- Example: Paracetamol

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FACTORS AFFECTING DRUG METABOLISM

There are various factors affecting drug metabolism as follows:

- Age
- Genetics
- Liver Health
- Gender
- Diet and Lifestyle
- Drug Interactions

① AGE

- Newborn babies shows slow metabolism as their liver is not fully developed.
- Liver functions decline with age, hence eldely people also shows slow metabolism.

② Genetics

- Some people naturally break down drugs faster or slower due to differences in their genes.

③ Liver Health

- If the liver is damaged (due to any liver disease), it cannot process drugs properly.

④ Gender

- Some drugs speed up or slow down metabolism when take together.

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⑤ Diet and Lifestyle

- Grapefruit Juice slows metabolism of some drugs, leading to higher drug levels.
- Smoking & Alcohol increase metabolism, making some drugs work less effectively.

⑥ Gender

- Men and women process some drugs differently due to hormones.

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EXCRETION

- Excretion is the process by which a drug and its metabolites are removed from the body primarily through urine, bile/ feces or other biological fluids.
- It is the final step in the process of drug elimination from the body.

IMPORTANCE OF EXCRETION

- Excretion is essential for:
  - Terminating drug action once its therapeutic purpose is fullfilled.
  - Preventing drug accumulation, which can lead to toxicity.
  - Help determining dosing interval & drug choice in patient with organ dysfunction.

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ROUTES OF DRUG EXCRETION

There are two major routes of drug excretion:

① Renal excretion
② Non-renal excretion

Renal Excretion Of Drugs

Renal excretion is the process by which drugs & their metabolites are eliminated from the body through the kidney via urine.

Steps In Renal Excretion

Renal excretion involves 3 main steps:

1. Glomerular filtration
2. Tubular secretion
3. Tubular reabsorption

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① GLOMRULAR FILTRATION

- Occurs in glomerulus.
- Only (free) unbound drug in plasma is filtered.
- Filtration rate depends on:
  - Renal blood flow
  - Glomerular filtration rate (GFR)
  - Drug size and binding.

② TUBULAR SECRETION

- Happens in the proximal tubule.
- Active process using transporters.
- Two types:
  - Organic ~~Aio~~ Anion Transporters (OAT) - for acidic drug (eg penicillin)
  - Organic Cation Transporters (OCT) - for basic drugs (e.g. morphine)
- Not affected by protein binding.

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③ TUBULAR REABSORPTION

- Passive process, mostly in distal tubule.
- Depends on:
  - Lipid solubility of the drug
  - Urine pH
  - Drug ionization (non-ionized = more reabsorbed)
- Weak acids are reabsorbed more in acidic urine; weak bases in alkaline urine.

Factors Affecting Renal Excretion Of Drugs

- Renal excretion involves three major processes : Glomerular filtration, tubular secretion and tubular reabsorption.
- Several physiological, pharmacological and pathological factors influence how efficiently drugs are excreted by the kidneys.

① Plasma Protien Binding

- Only free (unbound) drugs in plasma can undergo glomerular filtration.
- Highly protein bound drugs (e.g. warfarin, diazepam) have reduced filtration.
- However, protein binding does not prevent active tubular secretion, as the drug associates from protein to maintain equillibrium.

② Renal Blood Flow

- Increased renal blood flow increases glomerular filtration & delivery of drug to the kidneys.
- Reduced renal perfusion (e.g heart failure, shock, dehydration) decreases excretion.

③ Glomerular Filtration Rate (GFR)

- Drugs are filtered at the glomerular if not protein-bound.
- GFR determines the volume of plasma filtered per unit time (normal ~ 120 mL/min)
- Decreased GFR (e.g., in kidney disease, aging) reduce drug clearance and may lead to accumulation and toxicity.

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④ Active Tubular Secretion

- In the proximal tubule, drugs are secreted into the tubular lumen via specific transporters :

- Organic Anion Transporters (OAT) for weak acids (eg penicillin).
- Organic Cation Transporters (OCT) for weak bases (e.g morphine).

- This process is saturable and subject to competetion:
- E.g. probenecid inhibits secretion of penicillin, increasing its plasma level.

- Aging and renal impairment reduce active secretion capacity.

⑤ Tubular Reabsorption

- In the distal tubule, drugs may be reabsorbed back into the bloodstream by passive diffusion.
- Depends on :
  - Lipid solubility - Lipophilic drugs are reabsorbed more.
  - Ionization - Non-ionized drugs cross membranes easily.
  - Urine pH - Determines drug ionization state.
- Weak acids are reabsorbed in acidic urine (e.g aspirin).
- Weak bases are reabsorbed in alkaline urine (e.g. amphetamine).
- Reabsorption reduces renal excretion and prolongs drug action.

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⑥ URINE pH

- Affects ionization of week acids and bases, thereby influencing reabsorption.
- Manipulating urine pH can enhance a drug elimination in overdose:
- Alkanization (e.g, with sodium bicarbonate) promote excretion of weak acids.
- Acidification (e.g., with ammonium chloride) promotes excretion of weak bases.

⑦ AGE

- Neonates have immature kidneys → low GFR, tubular secretion, and reabsorption.
- Drug elimination is slower; dosing must be adjusted to avoid accumulation.
- Elderly indivisuals often have reduced renal function even if creatine is normal.
- Renal clearance decreases → longer half-life and risk of toxicity.

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⑧ DISEASE STATES

- Renal diseases (e.g. glomerulonephritis, chronic kidney disease) reduce GFR and secretion.
- Cardiovascular conditions (e.g. heart failure) reduce renal perfusion and thus excretion.
- Hepatic diseases can impair metabolism, altering the quantity of drug available for excretion.

⑨ Drug Interactions

- Co-administered drugs may compete for the same transporters, altering excretion :
- Example : Probenecid inhibits OAT, reducing renal clearance of penicillin.
- Some drugs can inhibit renal transporters and increase toxicity of co-administered drugs (e.g., cimetidine reduces secretion of creatinine).

⑩ Physiochemical Properties Of Drug

- Molecular size : Small molecules are filtered more easily.
- Lipid solubility : Lipophilic drugs are more likely to be reabsorbed.
- Ionization : Ionized (charged) drugs are less reabsorbed and more excreted in urine.

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Non Renal Routes For Drug Excretion

- Non-renal excretion refers to the elimination of drugs and their metabolites from the body via routes other than kidneys.
- These include billiary, pulmonary, salivary, sweat, lacrimal, mammary, and skin/ gastrointestinal (GI) mucosa excretion.

① Billiary Excretion

- Drugs are secreted into bile by liver transporters like P-gp, MRP, BCRP.
- Molecular weight > 300 Da and conjugated (glucoronide) compounds are preferred.
- Excreted into bile → Intestine → Feces.
- Some drugs undergo enterohepatic circulation and get reabsorbed, prolonging half-life.

Examples: Rifampin, Erythromycin, Estrogen, Digoxin.

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② Pulmonary Excretion

- Volatile substances and gases are excreted via lungs through passive diffusion.
- Depends on drug's vapour pressure, lipid solubility and respiratory rate.
- Rapid excretion allows fat recovery after inhaled anesthesia.

Examples: Halothen, Isoflurane, nitrous oxide, Alcohol.

③ Salivary Excretion

- Drugs enter saliva by passive diffusion or active transport.
- Salivary excretion may cause unpleasant taste or local irritation.
- Swallowed drug can be reabsorbed (entero - salivary cycling).
- Examples - Lithium, Phenytoin, Rifampin, Iodide.

④ Sweat And Sebaceous Glands

- Small, lipophilic drugs diffuse into sweat.
- Usually insignificant for drug elimination.
- Can cause skin irritation or affect body odor.
- Examples : Rifampin, Heavy metals (e.g. arsenic, lead), some antibiotics (traces).

⑤ Breast Milk

- Drugs passively difuse into breast milk (slightly acidic → favors basic drugs).
- Lipid - soluble, low molecular weight, and less protein - bound drugs transfer easily.
- May cause adverse effects in infants due to immature liver/ kidney function.
- Examples : Tetracyclines, Chloramphenicol, Lithium, opioids.

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⑥ GIASTROINTESTINAL (GI) TRACT

- Drugs may be directly secreted into intestinal lumen or excreted unchanged via feces.
- Also included unabsorbed oral drugs.
- Especially relevant for oral drugs with poor absorption or billiary excretion.
- Examples : Vancomycin (oral), neomycin, tetracyclines.

BIOAVAILABILITY

- We know Bioavailability as fraction of administered drug that reaches the systemic circulation in unchanged form.
- In Biopharmaceutics is discovered in terms of how dosage form & route of administration affect how much drug reaches systemic circulation.
- So here we can define Bioavailability as "Rate & extent to which the above active ingredient (drug) is absorbed from a drug product & becomes available at site of action or in systemic circulation.

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KEY ASPECTS

- It measures drug absorption & first pass metabolism.
- It is expressed in percentage.
- By IV Route, Bioavailability is almost 100%.
- For oral or other routes, bioavailability can be reduced by incomplete absorption, enzymatic degradation or first pass metabolism.

Objectives Of Bioavailability

- To assess how quickly and how much of the drug is absorbed into the bloodstream.
- To compare different formulations or routes of administration (e.g. tablet vs injection).
- To help in the development and optimization of effective dosage forms.
- To evaluate the impact of food, other drugs or physiological conditions on drug absorption.
- To generate data needed for regulatory approval of new or generic drugs.
- To ensure consistent therapeutic effects and minimize side effects.
- To study the extent of first pass metabolism in orally administered drugs.

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Absolute Bioavailability

- Absolute bioavailability is the measure of the extent to which a drug reaches the systemic circulation when given by a non-intravenous route (like oral or intramuscular), compared to an intravenous (IV) dose of the same drug.
- IV administration is used as the reference standard, because it delivers 100% of drug directly into the bloodstream – no absorption barrier.

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Significance :

- Helps determine how effective an oral or other non-IV dosage forms is compared to IV.
- Important during drug (new drug) development and clinical trial phases.

Formula :

$$Fa = \left(\frac{AUC_{oral}}{AUC_{IV}}\right) \times \left(\frac{Dose_{IV}}{Dose_{oral}}\right) \times 100$$

Where :

- AUC = Area under the curve (plasma drug concentration vs time)
- AUC reflects how much drug is present in systemic circulation over time.

Example:

A drug is given orally (100 mg) and IV (50 mg)
AUC from oral = 200 μg·h/mL
AUC from IV = 400 μg·h/mL

$$Fa = \left(\frac{200}{400}\right) \times \left(\frac{50}{100}\right) \times 100 = 25\%$$

Interpretation: Only 25% of the oral doses reaches the systemic circulation — rest is lost due to incomplete absorption or first-pass metabolism.

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Relative Bioavailability

- Relative bioavailability compares the bioavailability of a drug from one formulation or route of administration to another non-IV formulation or route of administration (reference) of the same drug.
- The reference is typically an established standard formulation (e.g. an oral solution, a different tablet formulation, or a different non-IV route).
- This is often used when an IV formulation is not available or feasible for comparison.

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Significance :

- Used in bioequivalence studies to compare a test formulation (e.g. generic) to a reference (e.g., innovator drug).
- Required for regulatory approval of generics.

Formula :

$$F_R = \left(\frac{AUC_{test}}{AUC_{reference}}\right) \times \left(\frac{Dose_{reference}}{Dose_{test}}\right) \times 100$$

Example :

- Test (generic) drug : 100 mg , AUC = 220 μg·h/mL
- Reference (brand) drug : 100 mg , AUC = 200 μg·h/mL

$$F_R = \left(\frac{220}{200}\right) \times \left(\frac{100}{100}\right) \times 100 = 110\ \%$$

Interpretation : The best formulation has 110 % relative bioavailability compared to the reference — it delivers slightly more drug to the bloodstream.

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Measurement Of Bioavailability

The methods useful in quantitative evaluation of bioavailability can be broadly divided into two categories :

1. Pharmacokinetic methods
2. Pharmacodynamic methods

Pharmacokinetic Methods

- Pharmacokinetic methods commonly used to quantify bioavailability using measurable biological samples (mostly blood or urine).
- It can be further subdevided into two types :

a. Plasma Drug Concentration Method
b. Urinary Drug Excretion Studies

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a. Plasma Drug Concentration Method

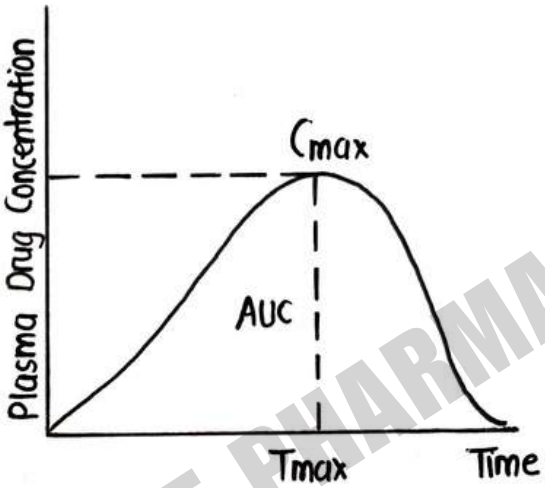
This is the most reliable and widely used method to measure bioavailability.

Principle

- When a drug is absorbed into the systemic circulation, it appears in the plasma.
- Measuring drug concentration in plasma over time helps determine both rate and extent of absorption.

Procedure

- The drug is administered (oral, IV route).
- Blood samples are collected at various time intervals.
- Plasma is separated and analyzed for drug concentration.
- A plasma concentration vs time graph is plotted.



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KEY PHARMACOKINETIC PARAMETERS

ⓐ Cmax -
- It is minimum Plasma Drug Concentration.
- Indicates rate of absorption.
- Higher Cmax = Faster absorption.

ⓑ Tmax -
- Time to reach Cmax
- Indicates how quickly drug is absorbed.

© AUC -
- It is Area Under Curve
- Reflects the total exposure of drug in plasma.
- Directly proportional to the extent of absorption.

Uses

- To compare two different formulations (test vs reference).
- To study bioequivalance.
- To optimize dosage form designe.

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ⓑ URINARY DRUG EXCRETION STUDIES

This method is based on measuring the amount of unchanged drug excreted in the urine over time.

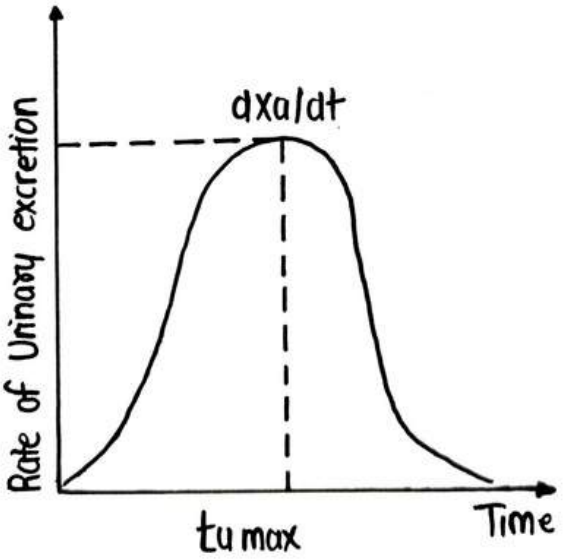
Principle

- Only the absorbed drug can enter the bloodstream and be filtered by the kidneys.
- Therefore, the amount of unchanged drug excreted in urine reflects the extent of drug absorption.

Procedure

- Administer the drug.
- Collect urine samples at regular intervals (e.g. every 2 hours).
- Analyze each sample for unchanged drug content.
- Plot : Rate of urinary excretion (dDu/dt) vs MidPoint Time (T).

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Key Parameters

- dXu/dt : - It is the maximum urinary excretion rate.
  - Gives an idea about rate of absorption.

- Tu max : - Time of maximum urinary excretion rate.
  - Similiar to Tmax in plasma studies.

- Xu : - Total amount of Drug Excreted in Urine
  - Similiar to AUC of Plasma Drug concentration studies

Conditions For Use

- The drug must be : - Excreted unchanged in urine.
  - Excreted is sufficient quantity for accurate measurement.

- Advantages : - Non-invasive compared to blood sampling.
  - Useful when plasma concentration measurement is difficult.

- Limitations : Cannot be used for drugs that :
  - Are extensively metabolized.
  - Are not excreted via urine.
  - Have very low renal excretion.

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PHARMACODYNAMIC METHOD

- When direct plasma or urine measurements are not possible, bioavailability can be assessed using the pharmacological effects (i.e. body's response to the drug).
- This method is useful when :
  - Drug concentration measurement is tough.
  - Drug acts locally (e.g. eye drops, skin creams)
  - No reliable analytical method is available.
- It has two main types :

① Acute pharmacological response
② Therapeutic response.

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① ACUTE PHARMACOLOGICAL RESPONSE METHOD

Principle

Bioavailability is estimated by measuring the immediate pharmacological effect of the drug, which correlates with its concentration at the site of action.

Examples

- Change in blood pressure after an antihypertensive drug.
- Pupil dilation after eye drops like atropine.
- Muscle contraction / inhibition in lab animals.

Limitations

- Response may not always directly correlate with concentration.
- Subjective variation in measurement.
- Tolerance or desensitization can affect results.

② THERAPEUTIC RESPONSE METHOD

Principle

Bioavailability is evaluated by comparing the clinical outcome or therapeutic effect of the drug.

Examples

- Pain relief with analgesics (e.g. paracetamol).
- Siezures control with antiepileptic drugs.
- Blood glucose reduction in diabetic patients after antidiabetic treatment..

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In Vitro Drug Dissolution Methods

- In vitro drug dissolution testing models are laboratory methods used to measure how quickly and how much of a drug dissolves from its dosage form (like a tablet or capsule) into a liquid, outside the body (i.e., in vitro).
- These tests simulate conditions in the human gastrointestinal (GI) tract and are important for predicting how the drug will behave in the body (in vivo).

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Purpose

- To ensure consistent drug release from batch to batch.
- To predict bioavailability and bioequivalence.
- To support formulation development and quality control.

Main types of In Vitro Dissolution Methods

- Usp Apparatus I (Basket Method)
- USP Apparatus II (Paddle Method)
- USP Apparatus III (Resiprocrating Cylinder)
- USP Apparatus IV (Flow - through cell)
- USP Apparatus V - VII

① USP APPARATUS I (BASKET METHOD)

- The tablet / capsule is placed inside a rotating basket submerged in a dissolution medium.
- Suitable for capsule and floating dosage forms.

② USP APPARATUS II (PADDLE METHOD)

- A paddle stirs the dissolution medium while the dosage forms rests the at bottom.
- Commonly used for tablets.

③ USP APPARATUS III (RECIPROCATING METHOD)

- Used for controlled - release formulations.
- The drug moves up and down between different compartments simulating GI conditions.

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④ USP APPARATUS IV (FLOW - THROUGH CELL)

- Medium flows through a cell containing the dosage form.
- Suitable for poorly soluble drugs or modified - release forms.

⑤ USP APPARATUS V-VII

- Less common, used for special dosage forms like transdermal patches or chewables.

Key Parameters Measured

- % drug released vs Time
- Dissolution rate
- T90 (time to release 90% of drug)

Why It's Important

- It's a predictive tool for in Vivo performance.
- Saves time and cost before clinical trials.
- Required by regulatory authorities for approval of generic drugs.

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In Vitro - In Vivo Correlation (IVIVC)

- In Vitro - In Vivo Correlation (IVIVC) is predictive mathematical model that describes the relationship between a drug's in vitro dissolution or release and its in vivo absorption (usually measured as plasma drug concentration or exposure).

Why IVIVC is Important

- It helps in predicting how a drug will behave in the human body using laboratory (in vitro) test.
- It can reduce the need for expensive and time-consuming clinical studies.
- Supports formulation development, bioequivalence testing, and regulatory approval, especially for modified-release formulations.

Key Concepts

① **In Vitro Data:** Usually comes from dissolution testing (e.g. % drug dissolved over time).

② **In Vivo Data:** Comes from pharmacokinetic studies (e.g., plasma concentration vs time curves).

③ **Correlation:** A mathematical link that allows prediction of in vivo behaviour from in vitro results.

Levels Of IVIVC (According To FDA)

① Level A (Point-to-Point Correlation):

- Strongest and most prefferred type.
- There is a direct relationship between in vitro dissolution time points and in vivo absorption time points.
- Example - 30% dissolved = 30% Absorbed.
- Allows full prediction of plasma drug concentration-time profiles.

② Level B (Stastiscal Moment Analysis):

- Compares mean in vitro dissolution time to mean residence/absorption time in vivo.
- Uses statistical moments, not point-to-point correlation.
- Less informative than Level A.

③ Level C (Single Point Correlation):

- Correlates one dissolution point (e.g., 60 min) with one pharmacokinetic parameter (e.g. Cmax or AUC).
- Simple but limited predictive value.

④ Multiple Level C:

- Improves on Level C by using multiple in vitro time points and correlating them with multiple PK parameters.
- Still not as powerful as Level A but better than single point.

EXAMPLE

- Suppose a sustained - release tablet releases 50 % of its drug in 2 hours (in vitro) and in a pharmacokinetic study, the same amount of drug is absorbed in 2 hours (in vivo).
- If this pattern holds across other time points, a level A IVIVC exits.

CONCLUSION

- IVIVC is a powerful tool in pharmaceutical development that connects lab results with real - world drug behaviour.
- A strong IVIVC helps save time, reduce costs, and supports faster regulatory approval.

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BIOEQUIVALANCE STUDIES

- Bioequivalance means that two medicines (usually a brand-name drug and its generic version) work same in the body.
- They release the same amount of active drug at same speed and have the same effect.
- Bioequivalance refers to the similiarity between two drug products in terms of:
  - Rate of absorption (how fast the drug gets into bloodstream).
  - Extent of absorption (how much of drug gets into bloodstream).

- When two drugs products are equivalent, it means -
  - They have comperable bioavailability (i.e. they reach similiar drug levels in the blood).
  - They produce similiar therapeutic effects and safety profiles.
  - They are usually tested in healthy volunteers using pharmacokinetic studies, which compare parameters like:
    * Cmax (maximum concentration of drug in the blood)
    * Tmax (Time to reach Cmax)
    * AUC (Area under the curve, which shows total drugs exposure)

- Regulatory agencies like the FDA or EMA accept two drugs ~~or~~ as bioequivalent if their Cmax and AUC values fall within 80% to 125% of each other.

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Example

If you take a 500 mg tablet of Brand A and a 500 mg tablet of Generic, B and both:

- Reach the same peak level in your blood.
- Take the same time to reach that level.
- Stay in your system for the same time.

The Brand A and B (Generic) are considered bioequivalent.

Need Of Bioequivalent Studies

These studies are essential for regulatory approval of generic drugs, ensuring they are just as safe and effective as the original product.

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BCS Classification Of Drugs

- The term BCS stands for Biopharmaceutical classification system,
- It is a scientific framework for classifying drugs based on their solubility & permeability.
- It helps in predicting the drug absorption & bioavailability, and it is widely used in drug development and regulation.

Class	Solubility	Permeability	Absorption	Example
I	High	High	Good	Metaprolol
II	Low	High	Variable	Nifedipine
III	High	Low	Variable	Cimetidine
IV	Low	Low	Poor	Furosemide

IMPERFECT PHARMACY

Class I

- Drug in this class are highly soluble in nature & can easily permeate intestinal membrane.
- They rapidly absorbed after oral administration.
- Exhibit good bioavailability.
- Drugs in this category are often considered ideal for oral delivery.
- Examples : Metaprolol, Propanolol.

CLASS II

- These drugs have poor solubility but can easily permeate the intestinal membrane.
- The bioavailability of these drugs can be improved by formulations that enhance solubility.
- These drugs are also suitable for controlled release developments.
- Examples : Ketaconazole, Ibuprofen, Phenytoin.

CLASS III

- Drug in these class have good solubility but having difficulty in permeating intestinal membrane.
- Changes in the integrity of gastrointestinal membrane, such as from disease or drug interactions, may affect bioavailability.
- Presence of food can alter absorption process.
- Example : Cimetidine, Atenolol, Acyclovir.

IMPERFECT PHARMACY

CLASS IV

- Drugs in these class have poor solubility & permeability.
- They have poor absorption & bioavailability.
- These drugs may require novel drug delivery system.
- Drug development in this categogary is a challenging process.
- Example : Furosemide.

Methods For Enhancement Of Bioavailability

- Enhancing the dissolution rate and bioavailability of poorly soluble drugs is a critical challenge in pharmaceutical development.
- Poorly solubility can limit drug absorption in the gastrointestinal tract, leading to subtherapeutic effects.
- Several formulation and technological approaches are used to improve these properties :

① Particle Size Reduction (Micronization / Nanonization)

Principle : Reducing particle size increases the surface area available for dissolution.

Techniques :
- Micronization using jet mills or ball mills.
- Nanonization using high-pressure homogenization or wet milling.

Advantage : Simple, effective for BCS Class II drugs.

Limitation : May lead to aggregation, poor flow properties.

IMPERFECT PHARMACY

② Solid Dispersions

Principle : Dispensering the drug in a water-soluble carrier improves wettability and reduces crystallinity.

Types :
- Eutectic Mixtures
- Solid solutions
- Glass solutions or Amorphous solutions.

Carriers : PEG, PVP, HPMC etc.

Advantages : Increase solubility, dissolution rate.

Challenges : Stability and scalability.

③ Use of Surfactants

Principle: Surfactants reduce surface tension and improve wetting and solubilization.

Examples: Sodium Lauryl Sulphate (SLS), Poloxamers, Tween 80.

Advantages: Enhanced dissolution and bioavailability.

Consideration: Surfactant concentration must be optimized to avoid GI irritation.

④ Salt Formation

Principle: Converting a poorly soluble drug into a salt form can improve solubility.

Criteria: Drug must have ionizable functional groups.

Examples: Aspirin (Acetylsalicyclic acid) into its sodium salt.

Limitations: Not all drugs are suitable for salt formation, stability issues.

IMPERFECT PHARMACY

⑤ Solid Lipid Nanoparticles (SLNs) & Nanostructured Lipid Carriers (NLCs)

Principle: Drugs are incorporated into lipid matrices at the nanoscale.

Benefits: Controlled release, enhanced bioavalability, biocampatibility.

Applications: Suitable for lipophilic drugs.

⑥ **Complexation**

Cyclodextrin Complexes :
- Cyclodextrins (e.g. β-cylodextrin) form inclusion complexes with hydrophobic drugs.
- Improve solubility, stability and taste.

Ion Exchange Resins :
- Used for taste masking and controlled release.

⑦ **pH Modification**

Principle : Using buffers or adjusting pH locally to increase solubility.

Examples : Adding Citric acid or sodium biocarbonate to modify microenvironment pH.

Application : Often used in oral suspensions or fast-dissolving tablets.

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⑧ **Amorphous Formulation**

Amorphous Drugs have higher solubility than their crystalline counterparts.

Techniques : Spray drying, Melt quenching.

Challenge : Stability issues due to tendency to recrystallize.

THANK YOU

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